

Gold ORB: the longer history

0.25 and 0.65 D1 ATR stops | Configured period: 1 January 2017 to 9 September 2026

Decision: retain both candidates for controlled range-filter and entry-cutoff tests. The added history supports continued research, but it also reveals another weak year and a data-quality question that must be resolved.

Candidate	Historical result	Main trade-off
0.25 ATR	\$232,001.55 ending balance; 38.35% CAGR; PF 1.312.	Higher return; 33.56% reported equity drawdown; 552-day balance recovery.
0.65 ATR	\$39,409.28 ending balance; 15.21% CAGR; PF 1.244.	Lower historical drawdown at 12.93%; lower return; 425-day longest recovery.

What changed with the extra history

The 694 added trades in 2017-2019 made money in aggregate for both stops, but both lost in 2019. The system has therefore encountered at least two losing calendar years: 2019 and 2023. Strong 2024-2026 results are not the only source of profit, yet the earlier edge is weaker.

Both runs contain 2,391 fully closed trades, with identical entry times, directions and entry prices. Only the saved stop multiplier differs. Every trade reconciles to the exported balance and profit totals.

The entry-time finding

The 06:05-08:59 entry window contains 1,732 trades (72.44% of the sample) and contributes 88.28% / 90.17% of net cash profit for 0.25 / 0.65. This motivates an early-cutoff experiment. It does not establish that excluding later trades improves future returns.

All times in the hourly analysis are the recorded broker-server times. The report includes all 24 hours, cumulative cutoffs, and hour-by-year and multi-period comparisons. Cash attribution is separated from risk-normalized performance.

Prepared for Abhishek | 19 September 2026. Source: the two supplied native MT5 exports. First actual trade: 7 April 2017. Last actual trading date: 8 September 2026. No range filter or entry cutoff was active.

Data integrity and test configuration

A matched experiment, with a historical coverage limitation

Check	Finding
Files	ORB_2017-2026_025.xlsx and ORB_2017-2026_065.xlsx; Sheet1 Settings, Results, Orders and Deals.
Account and symbol	XAUUSD; USD 10,000 deposit; leverage 1:100; ICMarketsSC-Demo; ORB_V2.
Schedule	03:05-06:05 range; 18:55 close; broker clock. Entry cutoff disabled.
Signal and stops	Touch entry; M5 signal timeframe; 0.05 signal-ATR buffer; D1 ATR(14) stop reference; 1% balance risk.
Filters	ATR range filter OFF; percentage range filter OFF; strong-candle filter OFF; one entry per day.
Reconciliation	4,782 trading deals per file plus one deposit; 2,391 reconstructed trades. Net, gross profit, gross loss and each deal balance reconcile to cents.
Matching	All 2,391 entry timestamps, directions and filled prices match; no duplicate deal IDs or overlapping/open positions found.
Recorded costs	Commission and swap both zero in every trading deal. This does not prove spread or slippage was zero.
Coverage concern	Configured start 1 January 2017; first entry 7 April 2017. History quality is reported as 96%. Cause of the initial absence of trades is unverified.

Do not interpret January-March 2017 as demonstrated market inactivity. Possible causes include missing history, indicator warm-up or other EA entry guards; the exports cannot distinguish them. Review the tester Journal and M1/D1 history before treating the early period as fully validated.

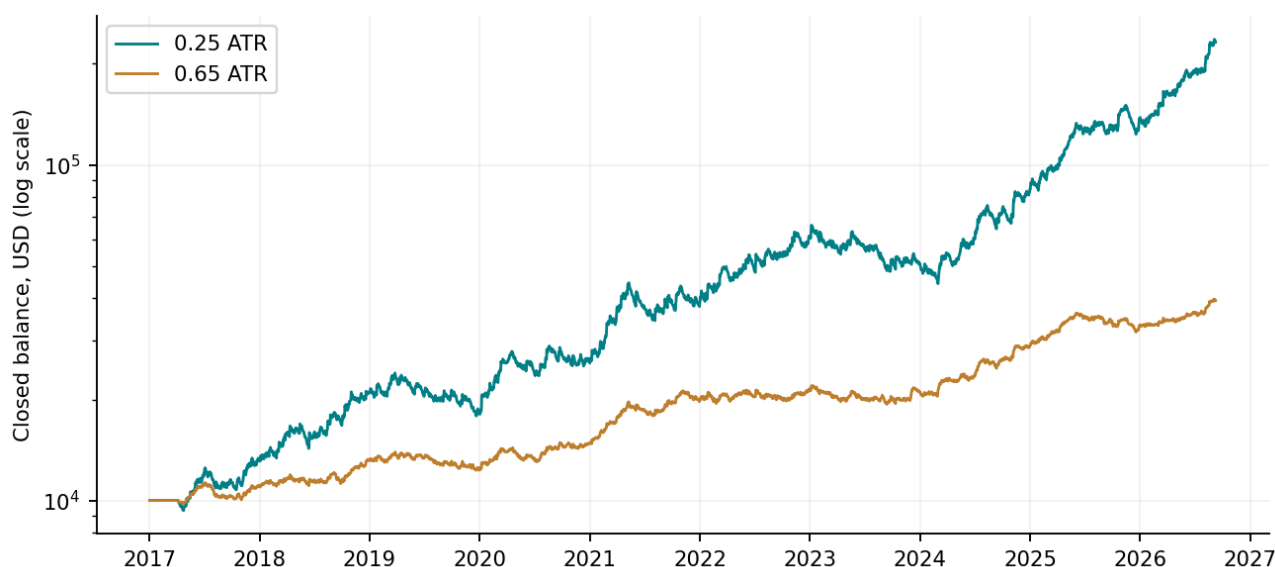
One exactly breakeven trade is separated from wins in each run. Strictly positive wins are 935 / 1,225, versus MT5 profit-trade counts of 936 / 1,226. Loss counts and monetary totals agree. The 2017 and 2026 annual results are partial trading periods.

The header chart timeframe M15 and Signal_Timeframe=5 (M5) serve different roles. The data-quality percentage alone does not certify real-tick execution or actual broker cost settings. No January-March trade history has been invented.

Returns and risk

Independent calculations on the reconstructed closed-balance series

Metric	0.25 ATR	0.65 ATR
Net profit, USD	222,001.55	29,409.28
Ending balance, USD	232,001.55	39,409.28
CAGR, configured span %	38.35	15.21
Strict win rate %	39.10	51.23
Cash profit factor	1.312	1.244
Mean budget R per trade	0.145	0.060
Budget-R profit factor	1.257	1.236
Maximum balance drawdown %	33.05	12.21
MT5 relative equity drawdown %	33.56	12.93
Daily Sharpe, zero risk-free	1.34	1.25
Daily Sortino, zero target	3.02	2.11
Annualized daily volatility %	25.75	11.42
CAGR / balance drawdown	1.16	1.25



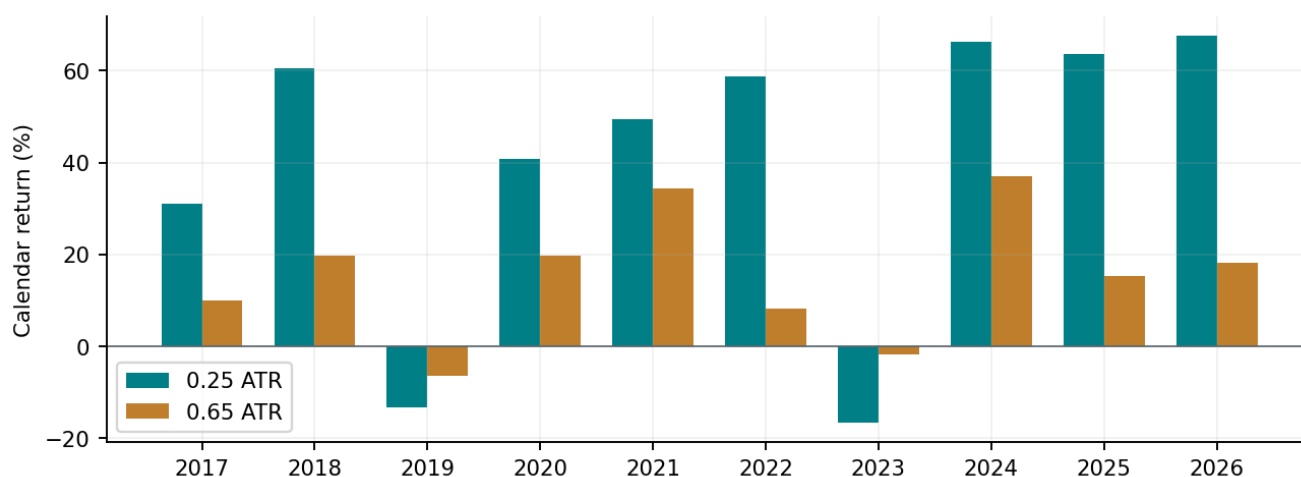
CAGR uses the configured 3,538-day span (ACT/365.25). From first actual entry to the configured end, CAGR would instead be 39.61% / 15.67%. Use the configured-span numbers consistently; the alternative is a coverage diagnostic, not a way to boost the headline.

Budget R = net trade P&L; / (1% of balance immediately before entry). Daily metrics use Monday-Friday closed-balance returns, including days with no recorded trades; this includes the unverified initial period. The chart is a balance curve, not an intratrade equity reconstruction.

Calendar-year comparison

Both variants lose in 2019 and 2023

Year	Trades	0.25 return	0.25 PF	0.65 return	0.65 PF
2017*	183	30.98%	1.275	10.00%	1.219
2018	258	60.56%	1.337	19.73%	1.288
2019	253	-13.20%	0.911	-6.49%	0.907
2020	251	40.73%	1.205	19.83%	1.279
2021	252	49.46%	1.217	34.27%	1.447
2022	256	58.83%	1.300	8.23%	1.102
2023	254	-16.68%	0.890	-1.65%	0.977
2024	258	66.36%	1.379	37.10%	1.551
2025	254	63.66%	1.337	15.26%	1.202
2026*	172	67.69%	1.693	18.24%	1.516



The 0.25 stop earns more in every positive year here, but loses more in the two negative years. In 2019 it returns -13.20% versus -6.49%; in 2023, -16.68% versus -1.65%. The wider stop changes the shape and size of losses; it does not eliminate weak regimes.

The additional early period has lower profitability: cash PF is 1.125 / 1.109 in 2017-2019, compared with 1.472 / 1.383 in 2024-2026. The higher full-history cash PF than budget-R PF also reflects capital growth weighting later trades more heavily.

* 2017 actual trades start in April; 2026 ends on 8 September. Annual returns are changes from each calendar period opening balance, not annualized partial-year returns. Eight positive rows out of ten include these two partial years.

Monthly and weekly consistency

Complete month-by-month return heatmaps; figures are percent

0.25 ATR monthly returns

2017	-	-	-	-4.1	13.4	9.0	-2.7	-1.7	-0.4	-1.7	16.3	1.3
2018	4.3	2.3	12.0	3.5	-4.4	1.0	2.2	11.2	-4.8	18.9	-0.5	5.0
2019	5.2	-7.6	13.9	-1.1	-6.8	-3.8	-2.4	-1.1	7.0	-7.1	0.1	-8.0
2020	16.4	6.7	13.3	1.0	-3.5	-3.6	4.2	12.2	-3.5	-6.6	3.3	-2.1
2021	7.7	21.7	9.2	10.3	-4.7	-2.5	-5.2	1.7	6.9	12.1	-10.6	-1.5
2022	13.7	-1.9	13.8	-4.4	6.0	-1.7	6.3	2.3	1.7	6.5	4.5	1.9
2023	-1.6	-3.8	-1.2	-1.3	8.7	-7.7	-1.7	-5.5	-2.6	-0.6	1.5	-1.4
2024	-3.8	-8.6	22.3	-4.7	9.6	6.2	20.4	-4.7	4.0	-1.5	13.5	4.6
2025	4.0	10.5	1.9	12.6	14.5	-0.1	-0.4	4.5	-2.1	11.6	-4.3	-0.3
2026	0.7	2.2	16.5	0.2	11.8	2.2	0.7	18.5	2.4	-	-	-
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

65 of 114 months with trades were positive. Best: 2024-03 (22.27%); worst: 2021-11 (-10.56%).

0.65 ATR monthly returns

2017	-	-	-	-1.1	7.4	4.1	-2.5	-3.5	-1.1	-1.8	6.3	2.5
2018	2.2	-0.4	2.7	-0.2	-0.6	-0.0	-0.8	5.8	-5.4	6.9	5.0	3.5
2019	2.6	-3.7	5.0	-0.3	-1.6	-0.3	-1.1	-3.5	2.6	-2.8	0.6	-3.8
2020	5.5	5.3	2.0	-0.7	-3.2	-0.2	3.5	2.9	3.0	-1.6	0.5	1.6
2021	4.9	9.5	3.8	6.2	0.9	-1.1	-2.8	2.3	6.2	6.3	-3.2	-2.0
2022	5.0	-5.0	4.2	-0.1	2.1	-2.8	0.4	-1.9	-0.2	0.4	2.6	3.7
2023	-2.0	-0.9	-2.0	-2.5	4.8	-4.5	-0.3	2.9	-2.1	-0.5	-0.1	5.9
2024	0.1	-2.4	11.9	-1.0	2.1	0.9	12.3	-1.8	0.9	2.8	6.5	0.9
2025	3.6	3.5	4.0	5.6	4.6	-1.0	-2.2	0.7	-2.3	1.6	-3.1	-0.3
2026	1.0	-0.3	3.0	-0.9	2.6	3.3	-0.7	9.2	0.1	-	-	-
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

62 of 114 months with trades were positive. Best: 2024-07 (12.26%); worst: 2018-09 (-5.37%).

Dash = no recorded trades before April 2017, or outside the test end; no inference of complete data coverage. September 2026 is partial. Color limits are symmetric at +/-15%; text retains actual values. Monthly returns compound within the period.

Rolling performance and difficult weeks

Consistency depends on the measurement horizon

Window*	ATR	Worst return	Best return	Positive windows
3 months	0.25	-22.85%	53.98%	71.59%
3 months	0.65	-9.63%	22.30%	67.05%
6 months	0.25	-18.36%	73.16%	82.72%
6 months	0.65	-9.74%	36.65%	76.97%
12 months	0.25	-24.77%	131.97%	87.87%
12 months	0.65	-6.69%	56.59%	83.96%

ATR	Positive active weeks	Best week return	Worst week return
0.25	262 / 493	14.81%	-5.03%
0.65	289 / 493	6.82%	-4.28%

0.25 ATR: five worst recorded weeks

Week starting	Trades	Return	Net USD
2023-02-06	5	-5.03%	-3,174.69
2018-06-04	5	-4.94%	-764.78
2019-12-02	5	-4.90%	-972.22
2022-01-10	5	-4.90%	-2,042.49
2023-09-04	5	-4.89%	-2,540.43

0.65 ATR: five worst recorded weeks

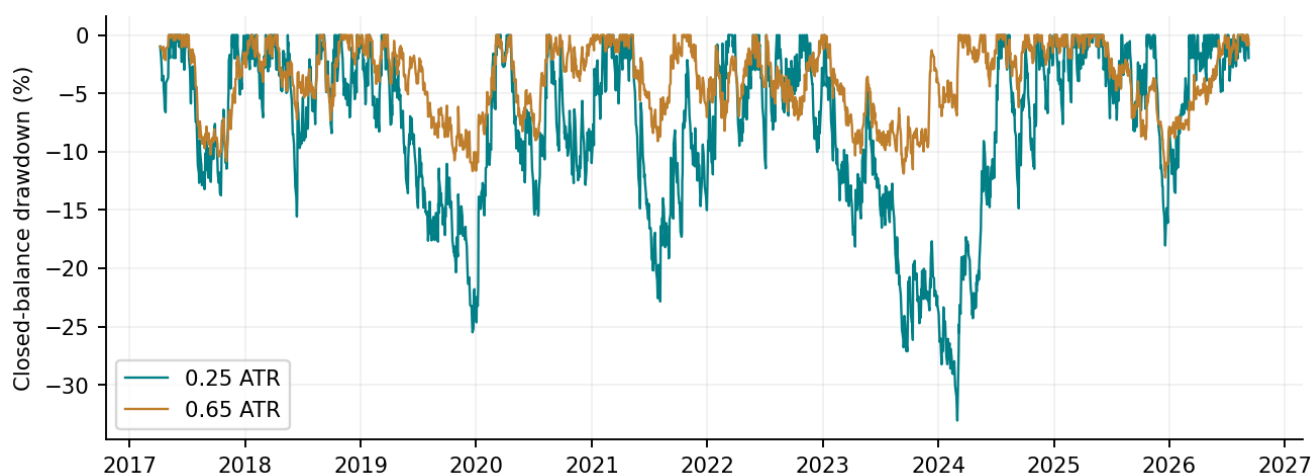
Week starting	Trades	Return	Net USD
2018-09-10	5	-4.28%	-514.98
2017-07-31	5	-3.43%	-370.30
2023-09-04	5	-3.41%	-689.94
2021-05-10	5	-3.26%	-639.33
2024-09-02	5	-2.92%	-757.63

A profitable long backtest can still contain many losing months and prolonged stagnation. The 0.25 variant has a worst rolling one-year return near -24.77%; 0.65 is materially less severe at -6.69%. A low positive-week fraction is compatible with a positive system when winners are larger than losers.

* Rolling windows use 63/126/252 weekdays as approximate 3/6/12 months. Overlapping windows are not independent samples. Weekly periods are Monday-Sunday (ISO-compatible week boundaries); partial first/last weeks remain included.

Balance drawdown and recovery

Recovery duration is a separate burden from drawdown depth



0.25 ATR: three longest peak-to-recovery episodes

Peak	Trough	Recovery	Days	Balance DD
2023-01-06	2024-03-01	2024-07-11	552	33.05%
2019-03-26	2019-12-20	2020-03-05	345	25.48%
2021-05-10	2021-08-05	2022-03-08	302	22.85%

0.65 ATR: three longest peak-to-recovery episodes

Peak	Trough	Recovery	Days	Balance DD
2023-01-06	2023-09-13	2024-03-06	425	11.88%
2025-06-05	2025-12-18	2026-06-25	385	12.21%
2019-03-27	2019-12-20	2020-03-05	344	11.67%

The 0.25 account remained below its January 2023 balance peak for 552 calendar days. Its 2019 episode lasted another 345 days. The 0.65 account also experienced lengthy recovery: 425 days from its January 2023 peak, and 385 days from June 2025.

These durations are measured from the last recorded closed-balance peak to the first recovered balance, including weekends. They are not estimated from an image. Intraday equity recovery dates cannot be independently reconstructed from these exports.

Current end-of-test unresolved episodes, if any, are retained in the analysis rather than discarded. The chart and episode table use closed balances; reported equity-risk figures appear in the headline metrics.

Trade behaviour and concentration

Both directions contribute, but return profiles differ

Metric	0.25 ATR	0.65 ATR
Wins / losses / breakeven	935 / 1455 / 1	1225 / 1165 / 1
Mean winner / mean loser, USD	999.31 / -489.59	122.42 / -103.48
Longest losing streak	17	12
Mean holding hours	7.75	10.29
SL exits / scheduled-close exits	1309 / 1082	385 / 2006
Worst single balance return	-1.20%	-1.02%
Largest winner / net profit	5.23%	3.47%
Top 5 winners / net profit	23.97%	15.03%
Top 10 winners / net profit	39.77%	26.34%
Mean R after removing top 10 R trades	0.111	0.047

ATR / side	Trades	Net USD	PF	Mean R
0.25 buy	1217	128,137.87	1.362	0.158
0.25 sell	1174	93,863.68	1.262	0.132
0.65 buy	1217	21,791.99	1.386	0.084
0.65 sell	1174	7,617.29	1.119	0.035

All positions close on the entry date. The 0.25 stop is hit on 1,309 trades; most of its payoff comes from the survivors held to the daily close. That is consistent with a low-win-rate, larger-winner profile. The 0.65 stop allows more trades to reach 18:55, including losing trades.

Top-ten cash winners account for 39.77% / 26.34% of net profit. After removing the ten largest normalized-R winners, mean R remains positive at 0.111 / 0.047. This is a diagnostic deletion exercise, not a simulated new strategy. Cash concentration is amplified by compounding.

Exit categories are determined after the result and cannot be used as entry filters. Short-side cash PF for 0.65 is only 1.119, so its margin over costs is narrower than the long side.

0.25 ATR: every entry hour

Full-history attribution | Broker-server clock | 00:00-23:59

Entry hour	N	Gross gain \$	Gross loss \$	Net \$	Cash PF	Mean R
00:00	0	0	0	0	-	-
01:00	0	0	0	0	-	-
02:00	0	0	0	0	-	-
03:00	0	0	0	0	-	-
04:00	0	0	0	0	-	-
05:00	0	0	0	0	-	-
06:00	889	386,183	305,324	80,858	1.26	0.123
07:00	445	197,083	130,331	66,752	1.51	0.201
08:00	398	153,790	105,426	48,365	1.46	0.246
09:00	327	75,151	77,434	-2,283	0.97	-0.042
10:00	150	64,805	40,663	24,142	1.59	0.282
11:00	70	23,588	22,885	703	1.03	0.080
12:00	29	7,447	7,296	151	1.02	0.191
13:00	19	5,138	6,996	-1,858	0.73	0.126
14:00	18	8,608	4,941	3,667	1.74	-0.084
15:00	24	8,957	3,264	5,693	2.74	0.573
16:00	14	2,327	7,452	-5,125	0.31	-0.248
17:00	7	1,238	340	898	3.64	-0.094
18:00	1	38	-0	38	-	0.391
19:00	0	0	0	0	-	-
20:00	0	0	0	0	-	-
21:00	0	0	0	0	-	-
22:00	0	0	0	0	-	-
23:00	0	0	0	0	-	-

Each label covers the hour up to xx:59:59. The 06:00 bucket starts at the permitted 06:05; the 18:00 bucket ends before 18:55. Empty hours are shown explicitly. Losses are positive magnitudes. Cash columns round to whole dollars here; calculations retain cents. PF is undefined with no losing trades, including the one-trade 18:00 bucket.

Assigning a trade to an hour uses its **entry** timestamp and the trade's final realized P&L, even if it closes much later. This table does not show the hour when profits accrued. Mean R removes most account-size weighting: 1 R means 1% of that trade's recorded entry balance.

06:00 provides the largest net dollars because it contains the most trades (889). The 08:00 bucket has stronger per-trade expectancy than 06:00. The 09:00 bucket has 327 trades and negative cash and normalized results. Large PF values after noon often come from very small samples.

0.65 ATR: every entry hour

Full-history attribution | Broker-server clock | 00:00-23:59

Entry hour	N	Gross gain \$	Gross loss \$	Net \$	Cash PF	Mean R
00:00	0	0	0	0	-	-
01:00	0	0	0	0	-	-
02:00	0	0	0	0	-	-
03:00	0	0	0	0	-	-
04:00	0	0	0	0	-	-
05:00	0	0	0	0	-	-
06:00	889	61,103	50,338	10,765	1.21	0.056
07:00	445	30,837	22,423	8,413	1.38	0.087
08:00	398	25,327	17,986	7,341	1.41	0.096
09:00	327	13,228	14,587	-1,359	0.91	-0.030
10:00	150	10,399	7,756	2,643	1.34	0.117
11:00	70	4,071	2,698	1,373	1.51	0.092
12:00	29	1,131	1,429	-298	0.79	0.025
13:00	19	838	882	-45	0.95	0.016
14:00	18	1,129	821	308	1.38	-0.001
15:00	24	1,378	740	639	1.86	0.152
16:00	14	404	855	-451	0.47	-0.115
17:00	7	106	40	66	2.66	0.017
18:00	1	15	-0	15	-	0.147
19:00	0	0	0	0	-	-
20:00	0	0	0	0	-	-
21:00	0	0	0	0	-	-
22:00	0	0	0	0	-	-
23:00	0	0	0	0	-	-

Each label covers the hour up to xx:59:59. The 06:00 bucket starts at the permitted 06:05; the 18:00 bucket ends before 18:55. Empty hours are shown explicitly. Losses are positive magnitudes. Cash columns round to whole dollars here; calculations retain cents. PF is undefined with no losing trades, including the one-trade 18:00 bucket.

Assigning a trade to an hour uses its **entry** timestamp and the trade's final realized P&L, even if it closes much later. This table does not show the hour when profits accrued. Mean R removes most account-size weighting: 1 R means 1% of that trade's recorded entry balance.

06:00-08:59 dominates contribution here too. The 09:00 bucket loses \$1,359.47 over 327 trades. At 12:00 and 13:00, cash results are negative while mean R is positive: later, larger-dollar trades can outweigh earlier smaller trades. Use both views when assessing timing.

0.25 ATR: hourly stability

Every year shown; cell value is mean budget R per entry

06:00	-0.30	0.12	-0.09	0.22	0.34	0.03	-0.14	0.23	0.26	0.22
07:00	0.24	0.34	-0.23	0.14	0.27	0.24	0.28	0.07	0.22	0.71
08:00	0.56	-0.09	0.08	0.10	0.40	0.87	-0.17	0.50	0.75	0.05
09:00	-0.05	0.33	-0.03	-0.26	-0.33	-0.23	-0.19	0.18	0.13	0.24
10:00	0.04	1.00	0.53	0.09	-0.38	0.94	0.47	-0.43	-0.65	1.96
11:00	0.12	0.32	-0.22	0.70	-0.39	0.04	-0.70	0.24	0.25	-0.36
12:00	0.65	-	-0.99	1.44	-0.44	0.03	0.59	-0.99	-1.00	0.15
13:00	0.92	0.08	-0.01	-0.14	-1.01	0.74	-0.99	1.67	0.04	-1.00
14:00	-1.00	-0.32	-0.15	-0.99	1.10	-	-0.99	-0.62	1.24	1.25
15:00	0.31	0.17	0.60	1.98	-1.00	0.51	1.01	1.87	0.32	0.08
16:00	-	-	-0.83	-	-	-	-0.34	0.75	-0.62	-1.00
17:00	-0.36	-	-0.45	-	-	-	-0.17	-	0.33	0.18
18:00	0.39	-	-	-	-	-	-	-	-	-
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026

Broader-period check: mean R (trade count)

Hour	2017-2019	2020-2022	2023	2024-2026
06:00	-0.012 (189)	0.184 (293)	-0.144 (104)	0.238 (303)
07:00	0.057 (104)	0.219 (168)	0.277 (42)	0.270 (131)
08:00	0.166 (153)	0.385 (109)	-0.167 (44)	0.412 (92)
09:00	0.055 (142)	-0.270 (94)	-0.188 (34)	0.177 (57)
10:00	0.510 (51)	0.213 (49)	0.471 (13)	-0.008 (37)
11:00	0.140 (21)	0.190 (18)	-0.696 (5)	0.105 (26)
12:00	0.419 (7)	0.306 (12)	0.593 (2)	-0.281 (8)
13:00	0.318 (6)	0.214 (5)	-0.994 (2)	0.236 (6)
14:00	-0.369 (7)	0.400 (3)	-0.994 (2)	0.309 (6)
15:00	0.364 (8)	0.676 (8)	1.006 (3)	0.485 (5)
16:00	-0.834 (2)	- (0)	-0.340 (2)	-0.112 (10)
17:00	-0.390 (3)	- (0)	-0.168 (1)	0.228 (3)
18:00	0.391 (1)	- (0)	- (0)	- (0)

The 07:00 hour is positive in each of these four broad periods for both stops. That makes it worth preserving in the first cutoff tests. It is not an independent confirmation: periods and hours have been examined using the same historical sample.

A dash in the heatmap means no trades, not zero expectancy. Later-hour cells can contain very few observations; a green cell does not establish stability. The 09:00 hour is positive again in 2024-2026, while 10:00 becomes much weaker there. Avoid designing a rule only to erase 2019 or 2023.

0.65 ATR: hourly stability

Every year shown; cell value is mean budget R per entry

06:00	-0.28	0.10	-0.05	0.11	0.24	-0.07	-0.00	0.13	0.08	0.05
07:00	0.12	0.10	-0.05	0.01	0.17	0.13	0.13	0.03	0.10	0.19
08:00	0.25	-0.08	0.02	0.09	0.15	0.22	-0.08	0.30	0.23	0.02
09:00	-0.00	0.02	-0.03	-0.11	-0.04	-0.13	-0.15	0.06	0.11	0.11
10:00	0.09	0.39	0.20	0.10	-0.32	0.31	0.30	0.12	-0.41	0.69
11:00	-0.13	0.22	-0.23	0.50	-0.01	0.49	-0.32	0.03	0.13	0.08
12:00	0.23	-	-0.49	0.53	-0.12	0.07	0.22	-0.96	-0.83	0.06
13:00	0.36	0.03	0.06	-0.05	-0.65	0.07	-0.37	0.62	0.17	-0.48
14:00	-0.99	-0.15	-0.10	-0.28	0.45	-	0.09	-0.08	0.35	0.47
15:00	0.05	0.07	0.23	0.66	-0.50	0.19	0.23	0.67	0.12	0.03
16:00	-	-	-0.34	-	-	-	-0.13	0.13	-0.16	-0.42
17:00	0.05	-	-0.18	-	-	-	-0.06	-	0.11	0.07
18:00	0.15	-	-	-	-	-	-	-	-	-
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026

Broader-period check: mean R (trade count)

Hour	2017-2019	2020-2022	2023	2024-2026
06:00	-0.003 (189)	0.078 (293)	-0.005 (104)	0.092 (303)
07:00	0.031 (104)	0.106 (168)	0.133 (42)	0.092 (131)
08:00	0.058 (153)	0.142 (109)	-0.077 (44)	0.186 (92)
09:00	-0.005 (142)	-0.096 (94)	-0.155 (34)	0.088 (57)
10:00	0.226 (51)	0.030 (49)	0.305 (13)	0.016 (37)
11:00	0.004 (21)	0.331 (18)	-0.318 (5)	0.077 (26)
12:00	0.125 (7)	0.144 (12)	0.222 (2)	-0.290 (8)
13:00	0.152 (6)	-0.098 (5)	-0.369 (2)	0.102 (6)
14:00	-0.254 (7)	0.208 (3)	0.093 (2)	0.159 (6)
15:00	0.094 (8)	0.168 (8)	0.228 (3)	0.173 (5)
16:00	-0.338 (2)	- (0)	-0.127 (2)	-0.068 (10)
17:00	-0.023 (3)	- (0)	-0.062 (1)	0.084 (3)
18:00	0.147 (1)	- (0)	- (0)	- (0)

The 07:00 hour is positive in each of these four broad periods for both stops. That makes it worth preserving in the first cutoff tests. It is not an independent confirmation: periods and hours have been examined using the same historical sample.

A dash in the heatmap means no trades, not zero expectancy. Later-hour cells can contain very few observations; a green cell does not establish stability. The 09:00 hour is positive again in 2024-2026, while 10:00 becomes much weaker there. Avoid designing a rule only to erase 2019 or 2023.

Entry cutoffs: full attribution table

All windows start at 06:05; end times are exclusive

End	N	0.25 net \$	0.25 sum R	0.65 net \$	0.65 sum R
07:00	889	80,858	109.02	10,765	49.63
08:00	1334	147,611	198.66	19,178	88.26
09:00	1732	195,976	296.65	26,519	126.27
10:00	2059	193,693	282.83	25,160	116.38
11:00	2209	217,835	325.11	27,803	133.93
12:00	2279	218,539	330.71	29,176	140.37
13:00	2308	218,690	336.26	28,878	141.09
14:00	2327	216,832	338.66	28,833	141.39
15:00	2345	220,499	337.14	29,141	141.38
16:00	2369	226,191	350.90	29,780	145.02
17:00	2383	221,066	347.43	29,328	143.42
18:00	2390	221,964	346.77	29,395	143.54
18:55	2391	222,002	347.16	29,409	143.69

These are sums of retained historical trades at their original sizes, not projected results of a filtered EA. Removing trades changes the subsequent balance and dynamic lot size. A new MT5 run is required for the actual filtered profit, drawdown and CAGR.

Before 09:00: 1,732 trades, 296.65 / 126.27 budget R. Before 12:00: 2,279 trades, 330.71 / 140.37 budget R. The full day has 347.16 / 143.69 budget R. Noon therefore retains 95.32% of trades and 95.26% / 97.70% of total normalized contribution.

The largest retained cash sum in this table happens at 16:00 for both stops, but it removes only 22 later trades. Selecting it because it is the maximum would be a fragile choice. The larger, economically simpler comparison is an early-only window versus a morning window versus the original full window.

A 09:00 cutoff also removes profitable 10:00 and 11:00 trades. A single contiguous window cannot exclude only 09:00-09:59 while keeping later hours. Do not describe a cutoff as an isolated test of the 09:00 hour.

All figures are before any missing cost adjustment. Later entry starts are harder to infer: V3 touch mode can enter immediately when price is already outside at the delayed start, producing trades absent from this original ledger.

Cost tolerance and actual sizing

Recorded commission is zero; costs below are hypothetical additions

Extra \$ / round-trip lot	ATR	Adjusted net \$	Cash PF	Mean R
0	0.25	222,001.55	1.312	0.145
0	0.65	29,409.28	1.244	0.060
7	0.25	210,753.25	1.293	0.132
7	0.65	27,511.23	1.226	0.055
20	0.25	189,863.55	1.259	0.108
20	0.65	23,986.28	1.195	0.046
27	0.25	178,615.25	1.242	0.095
27	0.65	22,088.23	1.178	0.042

The scenarios subtract the stated dollars times each recorded trade's volume. \$7 is an illustrative commission assumption; \$20/\$27 are broader execution-stress examples, not verified IC Markets charges. At an inferred 100-ounce contract per lot, \$20 per lot is equivalent to \$0.20 per ounce of additional round-trip price cost.

Both remain profitable under these fixed-volume scenarios, but that is not a complete execution test. Costs deducted during an actual run change balances, future lot sizes and potentially entry acceptance. Additional spread stress must avoid double-counting spread already embedded in the execution prices.

Sizing diagnostic	0.25	0.65
Mean planned stop risk % balance	0.991	0.949
Minimum planned risk %	0.938	0.521
Maximum planned risk %	1.000	1.000
Fixed-volume break-even extra \$ / lot	138.16	108.46

Both target 1% balance risk, but the actual rounded size averages about 0.991% / 0.949% planned risk. The wider stop sometimes has a small lot size and greater rounding drag. Its lower drawdown is therefore partly a sizing outcome, as well as a stop/exit outcome. No trade-count mismatch occurs here.

Planned stop risk is inferred from entry price, order SL, volume and the verified approximately \$100 profit per \$1 price move per lot. Actual fill losses can exceed planned risk; the worst 0.25 trade lost 1.20% of its entry balance.

Monte Carlo stress check

Resampling historical daily returns; conditional scenarios, not forecasts

ATR	Block days	Median DD	95th DD	99th DD	5th CAGR
0.25	5	28.64%	42.42%	50.60%	21.15%
0.25	20	28.18%	41.37%	48.27%	22.04%
0.65	5	15.11%	24.26%	29.81%	8.49%
0.65	20	15.44%	23.87%	28.05%	8.50%

For each stop and block length, 2,000 paths are generated by sampling circular blocks of 5 or 20 weekday returns with replacement. Each path has the original number of weekdays, compounds from a starting index of one, and includes the starting value when measuring peak-to-trough drawdown. Seed: 19092026.

Under these resamplings, the 95th-percentile closed-balance drawdown is about 41-42% for 0.25 and 24% for 0.65. These outcomes show why the observed historical drawdown should not be treated as a future ceiling. They also show that the wider-stop candidate is not immune to substantially deeper losses.

The simulation preserves short clusters within sampled blocks, but it breaks longer regime sequences. It assumes historical daily outcomes remain representative, includes the initial no-recorded-trade period as zero returns, and does not model new structural market changes, intratrade equity, new costs or broker lot rounding.

What this check can and cannot decide

It is a path-risk diagnostic for the observed return distribution. It is not an independent test of the strategy edge, a probability of passing a funded account, or an estimate of live ruin. No strategy-selection adjustment has been applied. A positive lower CAGR percentile does not establish a guaranteed positive future return.

The two daily-return series have correlation **0.825**. They share every entry and many adverse periods. Treat them as two risk profiles of one Gold ORB family when constructing the later portfolio.

The resampled drawdowns are based on closed balances. No full floating-equity series is available. Portfolio limits would require joint return paths, exposure and the applicable account rules.

The next tests to run

Range filter first; then a small, pre-recorded time-window experiment

1. Keep both unfiltered controls

Freeze these exports as the 0.25 and 0.65 controls. Save their input presets and tester settings. Resolve the 96% history quality and missing early trading period before interpreting fine differences as strategy improvements.

2. Range-filter batch: six new runs

For each stop	Minimum range / D1 ATR	Maximum range / D1 ATR
A: upper cap only	0.00	0.80
B: tighter upper cap	0.00	0.60
C: exclude tiny ranges too	0.10	0.60

Keep percentage filtering off, 1% balance risk, 03:05-06:05 range, 18:55 close and the original entry rules. Compare each filtered run with its own stop-matched control. These bounds are hypotheses, not fitted optima. Preserve the unfiltered variant even if a filter wins in-sample.

3. Time-window batch after selecting range variants

Entry start	Entry end	Question
06:05	09:00	Does the early concentration survive filtering and costs?
06:05	12:00	Can the morning window retain most contribution?
06:05	18:55	Original full-window control.

The hourly evidence adds 09:00 as a more informative first comparison than 15:00. All times are broker clock. Set `Use_Entry_Time_Filter=true` and the legacy `Use_Entry_Cutoff=false` in V3. Keep the actual position-close time at 18:55. These are entry cutoffs, not earlier exits.

Recompute hourly tables after the range filter: the eligible population may change. Evaluate calendar return, cost-adjusted mean R, drawdown duration, retained trades and stability across periods. Avoid choosing only the highest cash profit or tailoring the rule to one losing year.

Start with filter OFF in V3 and confirm the same baseline as V2. Delayed starts such as 07:00 should be a later, separately declared hypothesis because touch-mode entry prices and direction can change.

Decision and validation limits

Continue Gold development, then freeze the chosen rule set

0.25 ATR: stronger growth, larger recovery burden

Retain it as the return-oriented candidate. It has the higher aggregate and normalized payoff, positive early-period expectancy, contributions from both sides, and positive expectancy after removal of the largest ten normalized winners. Its lower win rate, 17-loss streak and long recovery periods remain material constraints.

0.65 ATR: lower historical variability, thinner payoff

Retain it as the lower-drawdown comparison. It reduces the depth of the two losing years and shows a slightly higher CAGR-to-drawdown ratio over the configured span. But its lower per-trade edge, weaker short side and long recovery times mean that a smoother curve is not sufficient evidence for live deployment.

Do not treat this history as unseen validation

The 2017-2019 extension is useful additional historical evidence if it was not previously inspected, but the full sample has now been reviewed for timing. Any rule selected from these hourly tables is in-sample for that decision. Reserve genuinely untouched data where available, or use a prospectively frozen forward test. Record all variants tried, including failures.

Inputs the future validation pipeline still needs

Available from these exports	Requires more evidence
Trade P&L; balance returns, entry hours, calendar results, costs recorded, realized drawdown and duration.	Tick/floating-equity history for intratrade risk; tester Journal for skip causes and coverage; verified spread, commission and slippage settings.
Stop-multiplier comparison and descriptive hourly attribution.	New MT5 runs for range/time rules and altered execution; untouched/forward data; parameter-neighbourhood runs; operational restart/close checks.

Finish the range and time comparisons, produce the finalist reports, then build the reusable Python validation pipeline. Use Pass / Warning / Fail / Insufficient data outcomes. A missing-input check must not silently pass. Freeze the final EA and parameters for forward testing before moving into ongoing live use.

No automatic live approval is given by sample count, profit factor or this report. Both variants belong to one strategy family; actual low correlation with a future second system must be measured on aligned returns and adverse periods.

Methods and source notes

Definitions sufficient to reproduce the principal calculations

Measure	Definition / scope
Trade reconstruction	Deals matched sequentially under verified one-position-at-a-time state. Entry order supplies SL. Match symbol, opposite direction, volume and chronology; exclude deposit and totals row.
Net and PF	Net = profit + commission + swap from entry and exit. Cash PF = sum positive net / absolute sum negative net. Zero outcomes classified separately.
Budget R	Net / (0.01 x balance immediately before entry). This is budget-normalized, not exact initial-stop R. R-PF uses positive/negative budget R sums.
CAGR	$(\text{Final balance} / 10,000)^{(365.25 / 3,538 \text{ days})} - 1$. Configured end is 9 September 2026; final trades occur 8 September.
Calendar returns	End balance / opening balance - 1. Period assignment uses exit date. Months/weeks with no recorded trades stay zero in numeric ledgers; unverified initial months are blank in heatmaps.
Daily ratios	Weekday closed-balance changes / preceding balance, zero risk-free and zero downside target. Sharpe = mean / sample SD x sqrt(252). Sortino denominator = RMS of min(daily return, 0), over all days.
VaR / expected shortfall	Loss magnitudes from empirical daily 5th percentile and average of returns at/below it. 0.25: 1.003% / 1.019%; 0.65: 0.976% / 0.989%. Historical estimates only.
Hourly and cutoff tables	Final trade outcomes grouped by entry broker hour. Cumulative sums retain original trade sizing. No claims of filtered CAGR or a new equity curve.
Drawdown paths	Closed-balance peaks include initial capital. Durations use calendar time. Equity headline is imported from MT5, not independently reconstructed.
Stress tests	Fixed-volume added-cost scenarios; circular moving-block resampling of weekday returns, 2,000 paths per block length. These have different assumptions from an EA rerun.

Primary evidence and references

Source exports: ORB_2017-2026_025.xlsx and ORB_2017-2026_065.xlsx. Sheet1: settings rows 4-60; results rows 62-82; orders start row 107; deposit row 4891; trading deals rows 4892-9673. Source row numbers, order and deal IDs were retained during reconstruction. No external market-price chart was inferred from these reports.

MetaQuotes, Testing Report: [official report metric definitions](#). Bailey et al., The Probability of Backtest Overfitting: [research on repeated strategy selection](#). These references support methodology, not the profitability of this EA.